



Full Length Article

Prediction of transportation systems energy demand using intelligent techniques

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ABSTRACT

Energy preservation and reduction of energy consumption is a major challenge faced by countries globally. By utilizing different Machine Learning (ML) models such as Stacking Ensemble Machine Learning (SEML) and Artificial Neural Network (ANN) algorithms, this paper aims to develop machine learning models to predict Transportation Energy Demand (TED) for different countries. Another major objective is to identify the most significant features that could help in predicting such information. Unlike the previous works that focused on specific countries, the proposed framework is generally applicable to many countries. Many experiments have been applied using benchmark dataset obtained from the Organization for Economic Co-operation and Development (OECD) in order to predict the TED for fourteen countries using available historical data from 2000 to 2020. The results showed that the SEML model performed better with an R2 of 0.9963 using the most significant feature list. Moreover, based on projections for 2040 data from recent research in Turkey, our study presents future predictions for Turkey's TED using our generalized framework. We estimate that TED will reach approximately 41 Million Tons of Oil Equivalent (MTOE) by 2040.

1. Introduction

Transportation is a fundamental component of urban development and a key enabler of smart city transformation. However, it is also among the most energy-consuming sectors worldwide, contributing significantly to environmental degradation through increased emissions and pollution. In addition, rising energy prices have made transportation systems economically challenging to sustain. Modern transportation and traffic management systems increasingly depend on advanced information and communication technologies to improve safety, efficiency, sustainability, monitoring, and control. In this context, Intelligent Transportation Systems (ITS) [1] have become central to urban planning and smart city initiatives, supporting advancements in road traffic management, traveler information services, autonomous vehicles, road safety, public transit efficiency, energy optimization, and environmental protection.

Energy consumption remains a critical challenge for many countries due to limited resources and escalating costs. Consequently, there is an urgent need for innovative research that leverages artificial intelligence techniques to address transportation energy-related challenges and

provide actionable insights [2–4]. Previous studies on TED prediction demonstrate considerable efforts using diverse methodologies and application contexts. However, most existing research focuses on individual countries, resulting in models and solutions tailored to specific national characteristics. While effective locally, such approaches often suffer from limited generalizability and may not perform well when applied to other countries with different economic, demographic, and transportation profiles. This limitation highlights the need for a generalized predictive framework capable of accommodating cross-country variations.

The objectives of this research are relevant to a wide range of countries, including Turkey, Canada, China, and others. The study aims to support transportation planning, reduce carbon dioxide emissions, facilitate the transition toward renewable energy sources, and assess existing energy strategies through accurate forecasting of TED across multiple countries. Accordingly, the proposed study aims to:

- Develop a generalized framework for predicting TED across multiple countries.

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- Incorporate novel features representing different transportation modes, including road and rail.
- Identify the most influential features contributing to TED prediction.
- Provide long-term TED prediction for a selected country, such as Turkey, up to the year 2040.

Based on these aims, the research is guided by the following central questions:

- **RQ1:** What are the most significant features that can be effectively utilized to predict TED?
- **RQ2:** How can a generalized predictive framework be developed to estimate TED across different countries?

Recent studies have used ML techniques to forecast TED, often focusing on country-specific models, notably in Turkey. While these models perform well nationally, their use is limited due to localized training data and a lack of cross-country applicability. This work presents a global SEML framework trained on multi-country data to address these limitations, using Turkey as a case study. Remarkably, even without being trained on Turkey's historical data, this model achieves predictive accuracy comparable to existing Turkey-specific studies. This analysis underscores the global framework's ability to capture TED dynamics across borders, providing a scalable alternative to conventional forecasting models.

This paper is organized as follows: first section is the background section, followed by a review of recent research in the related work section. Next, the proposed framework for predicting TED will be illustrated. Finally, results based on applying different ML models will be explained and a discussion section is presented before providing our final conclusion.

2. Background

2.1. Applying machine learning for ITS

The intricate and multi-layered architecture of ITS presents a valuable opportunity for the strategic deployment of ML techniques across various essential operational areas [5,6]. These areas encompass the physical infrastructure of road networks, the dynamic characteristics of vehicle fleets, and the behavioral patterns of system users. Furthermore, this includes the complex communication networks that facilitate connectivity, along with the unpredictable fluctuations in traffic flow. ML offers transformative solutions that have the potential to significantly improve system performance. This includes enhancing the accuracy of travel time predictions, developing detailed models of driver behavior to bolster safety and provide personalized assistance, optimizing adaptive control strategies for traffic signal networks, efficiently allocating and managing transportation resources, and fostering greater energy efficiency within the transportation sector. Advanced learning frameworks, particularly deep learning models built on multi-layered artificial neural networks, can be effectively utilized to address the intricate and high-dimensional challenges within the ITS landscape. Securing TED prediction requires data integrity and trusted deployment to ensure reliable forecasting. Beyond accuracy, implementing efficient cryptographic and hardware accelerated security is essential for processing data under resource constraints [7]. Such security-aware design is vital as intelligent transportation systems integrate predictive modeling into energy management [8].

The diverse and complex impact of ML on ITS can be comprehensively divided into three essential and interrelated areas of focus. Each of these task domains plays a crucial role in enhancing the efficiency, safety, and overall functionality of transportation systems through the integration of advanced algorithms and data-driven insights [5]:

- **Perception tasks** which aim to enhance contextual awareness through intelligent data interpretation. These tasks focus on the intelligent acquisition, accurate interpretation, and meaningful presentation of information obtained from diverse raw data sources. By utilizing advanced signal processing techniques, sophisticated computer vision algorithms, and effective sensor fusion methods, these tasks aim to accurately detect, identify, and classify relevant objects, dynamic events, and changing environmental conditions within the transportation context. For example, advanced object detection and tracking algorithms especially using deep learning (DL) have shown remarkable success in road sign recognition [9]. They autonomously learn hierarchical and invariant feature representations directly from raw pixel data, allowing for robust and precise identification even in challenging conditions, such as adverse weather or poor lighting. Additionally, these techniques are crucial for vehicle classification. They enable the detailed differentiation between various vehicle types [10], such as passenger cars, heavy goods vehicles, and motorcycles, based on their unique visual characteristics. This detailed classification is vital for numerous applications, including advanced driver-assistance systems, and traffic flow optimization strategies.
- **Prediction Tasks** which aim to anticipate future system states for proactive intervention. Given the inherently dynamic and often unpredictable nature of transportation systems, accurately forecasting future system states is crucial for proactive decision-making and minimizing potential disruptions. These tasks in ITS utilize a range of advanced time-series analysis techniques, robust regression modeling frameworks, and powerful neural network architectures. These methods help extrapolate future conditions based on a comprehensive analysis of historical trends and real-time data streams. For example, high-fidelity travel time predictions depend on modeling the temporal dependencies within historical traffic volume data. They also incorporate real-time congestion metrics from sensor networks, integrate external factors such as weather forecasts and incident reports. This approach provides users with highly reliable and context-aware travel time estimates [11], facilitating optimized route planning and reducing uncertainty for travelers. Moreover, ML is pivotal in addressing TED, which is the central focus of our research. In the upcoming sections, we will present comprehensive illustrations of this subject. In this context, advanced predictive models are developed to identify unusual patterns and accurately forecast future energy requirements. This capability facilitates the timely dissemination of critical information to stakeholders and aids in implementing proactive transportation management strategies aimed at reducing the impact of unpredictable events.
- **Management Tasks** which focused on the intelligent actuation and optimization of transportation system behavior. These tasks are designed to harness insights from ML to enhance the operational performance and overall efficiency of various components and sub-systems within ITS. By framing intricate transportation challenges as structured optimization problems, ML-based methodologies can autonomously identify and refine effective control strategies across a diverse array of parameters and real-time scenarios. For example, intelligent traffic light control systems [12] can utilize data-driven models that continuously engage with and adapt to the traffic environment. These models learn to dynamically modify signal timings based on observed traffic patterns, aiming to reduce congestion, shorten average travel times, and improve overall vehicle flow. In a similar vein, vehicle energy management systems [13], particularly in electric vehicles (EVs), stand to gain significantly from ML techniques that optimize driving behavior. By analyzing predicted traffic conditions, driver behavior patterns, and real-time energy consumption data, these systems can fine-tune operational parameters such as acceleration and braking strategies. Furthermore, forecasting short-term electricity demand is another challenge that requires an end-to-end solution to forecast electricity demand [14] for transportation in smart cities. This leads to enhanced energy efficiency,

extended vehicle range, and supports the broader objective of sustainable urban mobility.

The continual advancement of advanced ML techniques, coupled with the increasing availability of high-quality, multi-modal transportation data, presents a remarkable opportunity to develop smarter, more adaptive, and resilient ITS. The integration of ML into ITS is poised to bring about substantial enhancements in transportation safety, efficiency, sustainability, and overall user experience. This development paves the way for future mobility solutions distinguished by enhanced intelligence and responsiveness.

2.2. Transportation energy background

Transportation energy is crucial for the functioning of modern societies [15]. It is responsible for powering the movement of people and goods across vast distances. It encompasses various energy sources and technologies that are employed to propel vehicles, from gasoline and diesel to electricity and hydrogen [16]. In recent years, the pursuit of more sustainable and efficient transportation energy solutions has become increasingly important due to concerns over climate change, air pollution, and energy security. Innovations such as EVs [17], hybrid propulsion systems, and advancements in biofuels are changing the transportation energy landscape by offering cleaner and more sustainable alternatives to traditional fossil fuels. Furthermore, integrating renewable energy sources like solar and wind power into transportation infrastructure holds promise for reducing greenhouse gas emissions and mitigating the environmental impacts associated with conventional transportation energy sources. As societies strive for a greener and more resilient future, transforming transportation energy systems will play a critical role in shaping the sustainability and resilience of global transportation networks. In 2022, the transportation sector was the largest consumer of oil, accounting for 66% of the total consumption, according to Eurostat [18]. Road transport accounted for 48.6%, while air and water transport used 8.1% and 9.1% respectively [19]. In comparison to 2021, the total share was 63.3%, with road, air, and water transport using 48.7%, 9.1%, and 5.2% respectively [20]. This shows an increase of 2.7% in the transportation oil consumption shares between 2021 and 2022, highlighting the need for researches aimed at improving, developing, and advancing energy efficiency in transportation.

TED prediction in ITS depends on large scale sensing and distributed computation, where fault detection and secure processing are critical for reliable decision making. Prior studies on fault diagnosis and robust cryptographic hardware show how dependable computation can be sustained under faults and implementation level attacks, conditions common in transportation sensing environments [21,22]. Fault resilient cryptographic implementations further enable secure data exchange [23,24]. While this work focuses on improving ML based prediction accuracy, it aligns with these efforts by supporting resilient and trustworthy TED prediction pipelines.

The decision makers in several countries started to take decisions and strategies to reduce their country's overall energy consumption and limit the carbon dioxide emissions [25]. To achieve these goals in the transportation sector there are policies to use other renewable sources of energy, encourage purchasing and producing electric vehicles [26], and investing in the expansion and modernization of its public transportation systems, including buses, metros, and railways, to provide more efficient and sustainable alternatives to private car use. These investments aim to reduce congestion, decrease emissions, and improve the overall quality of urban transportation. Indeed, some governments around the world decided to reach zero emission vehicles transition by 2035 [27]. In Europe, the United Kingdom has brought forward its plan to end the sale of fully internal combustion engine (ICE) cars and vans to 2030, with a full transition to 100% zero-emission vehicle (ZEV) sales by 2035. Greece has strengthened its policy to only allow the sale of zero-emission light-duty vehicles (LDVs) from 2030 onwards. Switzerland

has surpassed its target of 15% electric car sales share by 2022, reaching 25% in 2022, and has set a new target of 50% electric car sales share by 2025. Italy has renewed its subsidy scheme with a focus on scrapping older, more polluting ICE vehicles, while Spain is focusing on scrapping, accompanied by support for lower-income individuals. Denmark is changing its taxation system to make EV company cars more attractive, and Finland has reduced import and annual taxes for EVs. In Canada, the federal government has increased its ambition for LDV deployment, with goals to achieve a zero-emission sales share of 20% by 2026, 60% by 2030, and 100% by 2035. British Columbia has also increased its target to 90% by 2030. In Australia, the government has funded a charging infrastructure program since 2020, and in 2022, it started providing competitive grant funding supporting the purchase of EVs for commercial use. New Zealand has set a target of 30% of the LDV stock to be ZEVs by 2035 and has introduced a purchase subsidy in mid-2021. In Japan, the Green Growth Strategy sets targets to reach 100% electrification of LDVs by 2035, and the 2023 Act on the Rational Use of Energy tracks and accelerates the targets set under the Strategy. In emerging markets and developing economies, governments are also increasing their ambition for EVs, with policies taking the form of tax exemptions, purchase incentives, mandates, and deployment targets. Indonesia has required government vehicles to be electric since 2022 and has introduced EV purchase subsidies from 2023. Seychelles aims for 30% of new vehicle sales to be electric by 2030, and Panama aims for 40% of the stock of selected public vehicles to be EVs by 2030. Vietnam has introduced a net-zero emissions target for the transport sector in 2050, with a ban on the production, assembly, and import of fossil fuel-powered vehicles in 2040. Ghana's recently released strategy contains a target that 4%, 16%, and 32% of new sales are to be EVs in 2025, 2030, and 2050, respectively. Additionally, countries recently proposing tax-related policies include Angola, Brazil, Ecuador, Pakistan, Trinidad and Tobago, Tunisia, Uzbekistan, and Vietnam [27].

This study introduces an original ensemble-based predictive framework for transportation energy demand that is designed to operate across multiple countries, overcoming the limitations of country-specific and single model approaches commonly reported in the literature. By integrating multiple learning models within a unified framework, the proposed framework improves robustness and generalization across heterogeneous transportation datasets. This scalable and transferable modeling strategy constitutes the main originality of the study.

3. Related work

Numerous studies have forecasted TED over upcoming decades for purposes such as planning, evaluating renewable energy adoption [28], assessing electric vehicle uptake, analyzing modal shifts for passengers or freight, and identifying the most influential features for decision-making. Researchers have achieved accurate TED predictions by comparing machine learning algorithms or developing novel approaches, including metaheuristics for complex optimization problems and hybrid algorithms combining multiple methods. Several studies also predicted associated CO₂ emissions, examining the relationship between transportation energy consumption and environmental impact. Transportation research relies on diverse datasets, both country-specific and global, with features typically categorized as socioeconomic (e.g., GDP, population), general (e.g., oil price), or transport-related (e.g., passenger, vehicle, and freight kilometers). Some studies further analyzed mode-specific features (road, rail, sea, air) and energy types (oil, gas, electricity, renewables), while others focused on aggregated feature values.

According to Korkmaz [29], transportation energy demand in Turkey was estimated for various modes using two metaheuristic algorithms: Bezier Search Differential Evolution (BeSD) and Black Widow Optimization (BWO). The dataset encompassed energy consumption, passenger-kilometers (passenger-KM), freight-kilometers (freight-KM), CO₂ emissions, and investments for different transportation modes in

Turkey (Highway, Railway, Seaway, and Airway) from 2000 to 2017. Data were gathered from a range of sources: the annual passenger and freight figures were obtained from the Turkish Statistical Institute (TSI) [30], carbon dioxide emissions data were sourced from the Ministry of Environment and Urbanization [31], while GDP and infrastructure investment information came from Eurostat statistics [32]. Energy consumption values were extracted from the balance tables of the Ministry of Energy and Natural Resources [33]. Mathematical models were formulated in four distinct forms: linear, exponential, power, and quadratic. The quadratic BeSD algorithm emerged as the most effective model, achieving the highest R^2 value of 0.9969. This model was subsequently utilized to forecast energy demand in Turkey for the year 2040 under two different scenarios. In the first scenario, the projection indicated that TED would reach 63 MTOE in 2040, based on estimates from the Ministry of Energy and Natural Resources. The second scenario proposed that a 5% demand were to shift from highway transportation to alternative modes, there could be an estimated 30% reduction in TED, resulting in a total of 42 MTOE by 2040.

The study by Tang, et al. [34] employs a data-driven multi-objective forecasting framework based on the Whale Optimization Algorithm (WOA) to predict transportation energy demand and CO2 emissions in Canada. The study evaluated several time-series and machine learning models, including Autoregressive (AR), Autoregressive Integrated Moving Average (ARIMA), Autoregressive Fractional Integrated Moving Average (ARFIMA), Seasonal Autoregressive Integrated Moving Average (SARIMA), Generalized Autoregressive Conditional Heteroscedasticity (GARCH), the Grey Model, Mixed Data Sampling (MIDAS), and Support Vector Regression (SVR), using multi-source transportation data (GDP, population, passenger numbers, rail freight volume, energy types, and CO2 emissions) collected from the World Databank [35], Statistics Canada [36], and the International Energy Agency [37] for the period 1990–2019. Results show that the WOA-based model outperformed individual predictors, achieving MAPE values of 0.003849 for energy demand and 0.003038 for CO2 emissions. The model projects that by 2048, transport energy demand will reach 3,835.59 PJ and CO2 emissions 288.04 Mt, with cumulative increases of 34.72–36.68% in energy demand and 30.04–50.0% in emissions compared to 2019. The analysis further indicates that a 5% reduction in oil, gas, and electricity consumption could reduce CO2 emissions by 4.8%, 0.315%, and 0.28%, respectively.

Hoxha, et al. [38] accurately predicted the TED in Turkey by using stacked ensemble models. They analyzed the historical data from 1975 to 2019, focusing on important features such as GDP, Population, Ton-KM, Vehicle-KM, Passenger-KM, Oil Price, and TED in MTOE. Data concerning GDP, population, Ton-KM, Vehicle-KM, Passenger-KM, and oil prices were obtained from the official website of TSI [30]. Energy consumption data for the transportation sector was sourced from the World Bank's 2020 report [39] and the World Energy Balances [40]. A total of nineteen machine learning algorithms were initially evaluated to determine the most suitable candidates for stacking configurations, including eXtreme Gradient Boosting. The ML algorithms used were Extra Trees Regressor (ETR), Random Forest Regressor, Gradient Boosting Regressor, Light Gradient Boosting Machine, ADABOOST Regressor, XGBoost, Orthogonal Matching Pursuit (OMP), Lasso Least Angle Regression, Ridge Regression, Linear Regression, Elastic Net, Least Angle Regression, Bayesian Ridge Regression, Decision Tree Regressor, Huber Regressor, K Neighbors Regressor, Passive Aggressive Regressor, and Dummy Regressor. The research is limited to Turkey's data and not compared to other countries. The authors conducted dimensionality reduction through permutation importance and hierarchical clustering based on Spearman rank-order correlations. Eight stacking ensemble models were constructed, employing four stacked algorithms with the Extra Trees Regressor and ADABOOST Regressor as *meta*-regressors. These models were assessed using six performance metrics: R^2 , MSE, MAE, RMSE, RMSLE, and MAPE. The ensemble model with OMP, BR, and LR as first-level algorithms and ETR as the *meta*-

regressor delivered the best performance in terms of R^2 for both training and test sets. They achieved an R^2 value of approximately 0.99 with all features and 0.98 with only two features. These results demonstrate the effectiveness of ensemble learning in predicting energy demand and its potential to guide sustainable transportation policies, aiding the shift toward a net-zero carbon economy.

Ägbulut [41] applied Deep Learning, Support Vector Machine (SVM), and Artificial Neural Network to forecast transportation-related CO2 emissions and energy demand in Turkey using historical data from 1975 to 2016, including GDP, vehicle kilometers, population, energy demand, and CO2 emissions. GDP, CO2, population, and energy data were sourced from the World Bank [39], while vehicle kilometers were obtained from TSI [30] and the Turkish General Directorate of Highways. Model performance was evaluated using R^2 , RMSE, MAPE, MBE, and MABE, alongside two regression-based mathematical models (linear and logarithmic) for projections to 2050. Model-II achieved superior performance, with R^2 values of 0.9360 for energy demand and 0.9140 for CO2 emissions, compared to 0.9285 and 0.9133 for Model-I. ANN and SVM generally outperformed DL, demonstrating high predictive accuracy overall. Projections indicate a 3.4-fold increase in both energy demand and emissions by 2050, with an annual growth rate of 3.65%, highlighting the need for policymakers to revise energy strategies to support sustainable economic development.

Sahraei, and Çodur [42] conducted a study in Turkey utilizing historical data from 1975 to 2019. This dataset encompassed various features, including year, population, oil price, GDP, passenger-kilometers, ton-kilometers, vehicle-kilometers, and million tons of oil equivalent. The data was sourced from multiple government agencies, such as the General Directorate of Turkish Highways, the World Bank [39], and the TSI [30]. The researchers developed eleven models to predict transportation energy demand. To achieve optimal results, they employed a hybrid algorithm that integrated Artificial Neural Networks with Genetic Algorithm (GA), Simulated Annealing (SA), and Particle Swarm Optimization (PSO). The ANN architecture comprised a single hidden layer, with weights and biases randomly generated for each hybrid model. Additionally, the features were normalized using a Min-Max scale ranging from 0 to 1. The study found that the hybrid model ANN-PSO, which incorporated three features (GDP, population, and ton-kilometers), yielded the most impressive performance, achieving an average R^2 value of 0.9962. These findings emphasize ANN-PSO as a versatile and robust tool for energy demand prediction, highlighting the critical need for targeted policies. Road transport continues to be the fastest-growing and least-regulated energy consumer, indicating that authorities should promote public transit, car-sharing, cycling, and invest in research and development for green energy transport. The authors recommend that future research explore additional swarm and nature-inspired optimization algorithms (e.g., glowworm, ant colony, bee colony, bat, spiral, and flower-pollination algorithms) and consider more granular variables such as vehicle type, road characteristics, and environmental factors.

Lashgari, et al. [43] projected Taiwan's transportation energy demand from 2019 to 2039 using an improved Emperor Penguin Optimizer (IEPO), based on 42 years of data (1974–2018) related to vehicle kilometers traveled, GDP, and population from the Taiwan Power Company and the Ministry of Energy and Natural Resources. The model analyzed various transportation modes up to 2018 and utilized metaheuristic algorithms for optimal forecasting with linear, exponential, and quadratic approaches. The study generated predictions based on population and GDP growth rates, yielding effective models for strategic energy planning in the Republic of China, although requiring significant computational resources. The RMS values during training for the linear, exponential, and quadratic models were 0.0452, 0.0461, and 0.0492, respectively, with R^2 values of 0.9491, 0.9367, and 0.9345. For testing, the RMS values were 0.0456, 0.0596, and 0.0642, with R^2 values of 0.9273, 0.9241, and 0.9154. The authors suggested developing an optimized fuzzy system to enhance future predictions.

Yao, Xu, and Ramezani [44] proposed a hybrid model combining Convolutional Neural Networks (CNN) with a variant of the Wildebeest Herd Optimizer (WHO), termed CNN/WHO, to predict long-term TED in Taiwan. The study utilized historical data from 1999 to 2019, including GDP, oil prices, population, passenger transport volume, and vehicle registrations. CNN/WHO performance was compared with multiple linear regression and a basic CNN, demonstrating superior accuracy. Results indicated that TED in Taiwan would remain nearly stable, at approximately 37.2 MTOE in 2020, despite GDP growth, aligning with Taiwan's green development strategy targeting CO2 reduction. Sensitivity analysis using projected GDP growth rates of 3%, 5%, and 7% suggested only a modest increase in TED from 35.8 MTOE in 2019 to 37.2 MTOE by 2025. The authors highlighted the need for continued CO2 reduction efforts and recommended exploring additional estimation methods, such as deep neural networks, to further enhance model performance.

While certain researchers have conducted analyses for various countries in their work, they approached each country as a distinct entity, developing a separate model for each to predict the TED. Maaouane, et al. [45] have utilized forty neural networks to predict energy demand in developing countries, leveraging the ODYSSEE database [46] which includes over 180 indicators for European countries [47] that aimed at monitoring and evaluating annual energy efficiency trends and energy-related CO2 emissions across various sectors, helping to inform policy decisions. The models were trained on the European data and subsequently evaluated using data from Morocco as a case study. The models were trained on European data and evaluated using Morocco as a case study. The dataset covered 28 European countries over the period 1990–2019 and included transport-related indicators such as average occupancy, average vehicle kilometers, average load, fuel economy, and stock vehicles, together with socio-economic variables including GDP, population (P), gasoline and diesel pump prices, price index, purchasing power parity, household final consumption expenditure, and population density. The majority of sectoral models achieved R2 values exceeding 90%, with exceptions observed in road transport for Malta (86.3%) and rail transport for Sweden (86.5%). All air transport models reported R2 values above 95%, while water transport performance was lower for Luxembourg (87.5%) and Lithuania (83.7%). Based on these models, four long-term scenarios were developed to project Morocco's energy demand in 2050. The "frozen efficiency" scenario, assuming no efficiency improvements, projected demand exceeding 10.5 MTOE, corresponding to a 74% increase relative to 2018. The second scenario, aligned with EU Commission recommendations, achieved up to 30% energy savings through improved fuel economy. The third scenario examined the adoption of electric vehicles, reducing consumption to 7.2 MTOE, which is 19% lower than the second scenario. The fourth "modal shift" scenario, emphasizing public transport and rail freight, yielded the largest reduction, with a 61% decrease compared to the frozen efficiency scenario. Additionally, substituting diesel with biofuels in buses and light vehicles reduced greenhouse gas emissions by up to 79%. The authors concluded that expanding public transportation, improving rail freight systems, and promoting biofuels and electric vehicles are critical for reducing energy demand, while emphasizing that model reliability depends on data quality and assumptions, necessitating regular updates and more comprehensive datasets.

Unlike earlier researches that employed various intelligent techniques, including machine learning, deep learning, hybrid methods, and metaheuristics for specific countries, to the best of our knowledge, there has been no existing work that has developed as a generalized model that could be applicable to multiple countries. Therefore, the proposed framework is developed and evaluated using data from multiple countries, thereby providing a more comprehensive analysis of TED. This framework facilitates the identification of both common patterns and trends across different countries, as well as features that are unique to specific contexts. As a result, the applicability and relevance of our

findings are significantly enhanced across a broader range of TED contexts.

This section provides a comprehensive review of recent studies in the relevant literature. Table 1 systematically summarizes these studies, highlighting their titles, research objectives, employed algorithms, dataset characteristics, temporal coverage, and key findings.

4. Proposed framework for predicting TED

The proposed framework is organized into a series of well-defined phases, as illustrated in Fig. 1. Starting with the data preparation phase, which involved meticulous collection and preprocessing of data to ensure its quality and relevance. During this step, data is cleaned, handled missing values, and transformed into a suitable format for analysis. Next, the feature engineering phase is applied by identifying, classifying, and categorizing the features that would be vital for machine learning models. This process involved rigorous analysis and selection of key variables to enhance the models' predictive power. The next phase entailed the applying of various machine learning algorithms. We carefully developed these models to leverage the engineered features and optimize their performance. Finally, during the evaluation phase, where we rigorously assessed our models using a comprehensive evaluation matrix. This included metrics such as R2, MSE, RMSE, MAE, and MAPE. This systematic approach allowed us to quantify the accuracy and effectiveness of our models, providing insight into their predictive capabilities.

4.1. Data preparation

4.1.1. Data collection

The dataset used for this study was collected from the historical data available in the Organization for Economic Co-operation and Development (OECD) [48]. The OECD is an international organization that strives to develop better policies for a better quality of life. Their goal is to shape policies that promote equality, prosperity, opportunity, and well-being for all. They draw on their 60 years of experience and insights to better prepare for the world of tomorrow. The organization has 38 member countries. The collected data [49] consists of nine predictors related to the Freight-KM and Passenger-KM for both road and rail transportation, infrastructure investments for road and rail transportation, GDP, Population, and Average-Oil-Price which collected from the U.S. Energy Information Administration (EIA) [50]. The target variable is the transportation fuel energy. The selection criteria for this study were to collect data from countries that had full records starting from 2000 till 2020. As a result, the model was trained on the historical annual data of 14 countries, including Sweden, Switzerland, Czech Republic, Spain, Finland, France, Hungary, Norway, Poland, Japan, United Kingdom, Slovak Republic, Australia, and Russia, from 2000 to 2020. Additionally, available data from Turkey was used to validate the model.

4.1.2. Data preprocessing

Before start working there are some steps required during the data preprocessing phase which is considered while the model development as follows:

- Neglect records with Null values
- Working on countries which have complete continues records.
- Working on the historical data of 14 countries Sweden, Switzerland, Czech Republic, Spain, Finland, France, Hungary, Norway, Poland, Japan, United Kingdom, Slovak Republic, Australia, and Russia from 2000 to 2020.
- We categorized the dataset features in order to examine different combinations.
- The models were evaluated using this matrix [R², MSE, RMSE, MAE, MAPE]
- Use the data of Turkey as a validation set to evaluate the model.

Table 1

List of publications.

Title	Purpose	Algorithms	Dataset Features	Dataset Years	Results
(Korkmaz 2023) [29]	Estimate the transportation energy consumption in Turkey	BeSDand BWO	MTOE, GDP, Passenger-km, Freight-km, CO2 emission from transport, Infrastructure investment per transportation mode	2000–2017	BeSD R ² : 0.9969
(Tang, et al. 2023) [34]	Establish a generalized framework that is capable of both energy demand and CO2 emissions forecasting in the transportation sector in Canada.	WOA andML Methods: AR, ARIMA, ARFIMA, SARIMA, GARCH, MIDAS,SVR	GDP, population, energy consumption (oil, gas, electricity, renewable, heat), and CO2 emissions	1990—2019	WOA TED MAPE: 0.003849 CO2MAPE: 0.003038
(Hoxha, et al. 2023) [38]	Predict transportation energy demand in Turkey	Eight different stacking ensemble models	GDP, population, vehicle-km, passenger-km, ton-km, oil price, energy demand	1975—2019	OMP, BR, LR with ETR R ² : 0.9891 (two features) R ² : 0.9955(All features)
(Āgbulut 2022) [41]	Forecast the transportation-based-CO2 emission and energy demand in Turkey.	DL, SVM,and ANN	GDP, Year, Population, vehicle-km, transportation energy demand, transportation-based CO2 emission	1970—2016	TED R ² : 0.9360 CO2 R ² : 0.9140
(Sahraei and Çodur 2022) [42]	Propose a hybrid algorithm, to better optimize the coefficients for predicting the energy demand in Turkey	ANN-GA, ANN-SA,and ANN-PSO	GDP, Year, Population, Oil price, Passenger-km, Ton-km, Vehicle-km, MTOE	1975—2019	ANN-PSO R ² : 0.9962 GDP, Population, ton-km R ² : 0.9491
(Lashgari, et al. 2022) [43]	Forecasting of the future transportation energy demand in Taiwan	IEPO	GDP, vehicle-km, population, gross rate	1974—2018	
(Yao, Xu and Ramezani 2021) [44]	Transport energy forecasting in Republic of China (Taiwan)	CNN/WHO	GDP, the cost of oil, individual,passenger transport amount, number of vehicle registrations	1999—2019	R ² : 0.9884
(Maaouane, et al. 2022) [45]	Develop a general model capable of learning from energy efficiency indicators of 28 European countries over a period of 30 years.	ANN	GDP, population, Pump Price Gasoline, Pump Price Diesel, the Price Index, Purchasing Power Parity, Household Final Consumption Expenditure, and Population Density.	1990—2019	The majority for each country R ² for road >= 0.90 R ² for rail >= 0.95 R ² for air >= 0.95 R ² for water>= 0.90

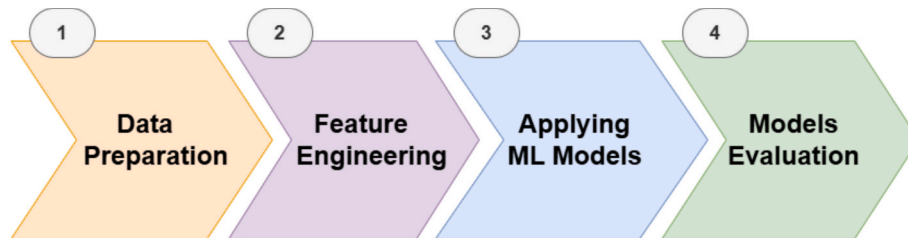


Fig. 1. Overview of the proposed machine learning framework, illustrating the sequential phases from data preparation to model evaluation.

- For the IND-ENE-FUEL-PC, we multiplied the amount of fuel by the population and divided it by 1 million to get the TED amount of fuel in tons per million units (MTOE).

4.2. Feature engineering

The collected dataset comprises various parameters, each addressing specific criteria related to either individual countries or global issues. In this research, we divide these features into two main categories: country-based features and global features in order to study the effect of globalization. This allows us to pinpoint the most significant features of each category. Next, evaluate each combination and its influence on our findings. After several rounds of analysis, we distinguished the top five feature combinations that emerged prominently. This will be explained in details in the experimental section.

4.2.1. Country-based features

The country-based features are further divided based on their focus: either transport or socioeconomic features.

Transport Indicators:

The effective use of these indicators necessitates several technologies for accurately identifying values, as they rely on Automatic Vehicle Location (AVL) [51] and Automatic Passenger Counting (APC) [52] systems. AVL typically employs Global Positioning System (GPS) technologies that are integrated with wireless communication networks to monitor vehicle movements and relay data to central dispatch centers. Conversely, APC systems utilize various sensors, including infrared beams, video analytics, head-counting cameras, or weight-sensitive floor mats, to track the number of passengers boarding and alighting at each stop. The successful implementation of these systems requires robust GPS coverage, reliable mobile networks (4G/5G), and cloud-

based platforms for data processing. Vehicles must be equipped with onboard computers, GPS receivers, and APC sensors. Moreover, data analytics capabilities are vital for effectively leveraging AVL and APC data. To ensure compatibility among transit operators, national or regional authorities should adopt open data standards and provide integration guidelines. Ultimately, a coordinated investment in both hardware and digital infrastructure is essential for the effective deployment of AVL and APC systems, which are often insufficient in many developing countries.

- T-GOODS-RD-TOT (Tonne-kilometre by road) in short TON-KM-RD: unit of measurement of goods transport which represents the transport of one tonne by road over a distance of one kilometre.
- T-GOODS-RL-TOT (Tonne-kilometre by rail) in short TON-KM-RL: unit of measurement of goods transport which represents the transport of one tonne of goods by rail over a distance of one kilometre.
- T-PASS-RD-TOT (Passenger-kilometre by road) in short P-KM-RD: unit of measurement representing the transport of one passenger by road over one kilometre.
- T-PASS-RL-TOT (Passenger-kilometre by rail) in short P-KM-RL: unit of measurement representing the transport of one rail passenger by rail over a distance of one kilometre.
- IND-INFR-RDINV-PC in short INV-RD: Road infrastructure investment in constant USD per inhabitant
- IND-INFR-RLINV-PC in short INV-RL: Rail infrastructure investment in constant USD per inhabitant

As ITS increasingly relies on data from connected vehicles and distributed sensing, emerging security attacks threaten data integrity and prediction reliability. Recent post-quantum cryptographic studies highlight fault-resilient authentication mechanisms, such as reliable hash trees for stateless signatures, to enhance robustness against implementation-level attacks [53]. Moreover, efficient hardware implementations of supersingular isogeny-based key exchange on FPGA and ARM platforms demonstrate secure communication under the latency and energy constraints typical of intelligent transportation systems [54,55]. Although cryptography-focused, these advances are directly relevant to securing transportation data pipelines that support trustworthy energy demand prediction.

Socioeconomic Indicators:

This category values are well-known for each country, and many international organizations, such as the World Bank [56], are interested in collecting this data.

- GDP-MLN-USD in short GDP: The Gross Domestic Product, commonly known as GDP, is a standard measure used to determine the value of goods and services produced in a country during a specific period. It takes into account the total income earned from that production, as well as the amount spent on final goods and services, excluding imports [57].
- POPULATION in short POP: The population of the country [58].

4.2.2. Global features

This category reflects data related to the global market rather than country-based parameters.

- OIL-PRICE-AVG in short OIL-AP: Average oil price per barrel in the year [50].

4.2.3. Dependent variable

Represents the TED, which is the target variable for our prediction.

- IND-ENE-FUEL-PC (TED): is the amount of motor fuel deliveries per inhabitant, measured in tones. Motor gasoline, which is a light hydrocarbon oil used in internal combustion engines (excluding those in aircraft), is included in this measurement.

4.3. Applying SEML approach

Ensemble learning [59] is one of the most powerful techniques in machine learning. It combines the output of two or more models/weak learners to solve computational intelligence problems, achieving higher performance. Stacking is a popular ensemble modeling technique in machine learning. Its ensembles various weak learners in parallel, and by combining them with *meta*-learners, we can make better predictions for the future.

In the stacking ensemble models [38], it was employed to predict the TED limit specifically for the Turkey dataset, achieving R^2 value of 0.9955. After evaluating a variety of machine learning algorithms, the authors identified the top eight models, which included four stacked algorithms alongside two *meta*-regressors. The proposed framework utilizes a Stacked Ensemble Machine Learning (SEML) models consisting of two levels [59]. In the first level, several machine learning algorithms are implemented including Orthogonal Matching Pursuit (OMP) [60], Bayesian Ridge Regression (BR) [61], Linear Regression (LR) [61], Lasso Regression [62], and Ridge Regression [63]. The second level employs either the Extra Trees Regressor (ETR) [64] or the ADABOOST Regressor [65] as *meta*-regressor.

In this research, we continued utilizing the eight models that demonstrated the best performance in [38]. However, instead of using data from a specific country, we extended the input to include datasets from multiple countries to develop a more generalized model for predicting TED. Furthermore, several features have been included described in section 4.2. As illustrated in Fig. 2, the proposed framework architecture is structured in two levels. The first level consists of four combinations of base regressors: [OMP, BR, LR], [BR, LR, Lasso], [OMP, BR], and [Lasso, Ridge, LR]. At the second level, either an ETR or AdaBoost is employed as the *meta*-regressor. To mitigate the risk of overfitting, a 4-fold cross-validation approach was applied, as recommended in [38]. Additionally, the number of estimators for the ETR was set to 100.

We adopted 4-fold cross-validation to ensure comparability with prior work [38] that employed the same setup. While 5-fold or 10-fold cross validation are more commonly used, 4-fold provided a suitable balance between computational efficiency and reliable performance estimation for our dataset, allowing robust evaluation without compromising comparability.

Extra Trees Regressor (ETR) [64] is an ensemble ML approach that involves training multiple decision trees and combining their results to make predictions. It is a robust and efficient algorithm suitable for a wide range of regression problems, including those in transportation and energy demand. This ensemble learning algorithm is based on the decision tree algorithm and builds several trees using random subsets of features and data points. The final prediction is determined by averaging the predictions of all individual trees, which helps reduce the model's variance and improves its predictive capability. Before using the Extra Trees Regressor, users need to specify the number of trees in the ensemble, the maximum depth of each tree, and the minimum number of samples required to split a node. The program then uses a random portion of the data to train the ensemble of decision trees and makes predictions by averaging the individual trees' predictions.

ADABOOST Regressor [65] is an ensemble learning technique specifically designed for regression tasks. It combines multiple weak learners, such as decision trees, to form a robust predictor. The fundamental concept behind ADABOOST involves training the weak learners sequentially, with each subsequent learner focusing on rectifying the errors made by the preceding ones. In order to emphasize the most challenging instances for the next learner, greater weights are assigned to the examples misclassified by the previous learners. It has a simple implementation that requires minimal hyperparameter tuning.

Orthogonal Matching Pursuit (OMP) [60] is a method used to represent sparse signals and learn dictionaries. It falls into the category of greedy algorithms, making a series of locally optimal decisions to find the best overall solution. The main idea is to express a signal as a linear

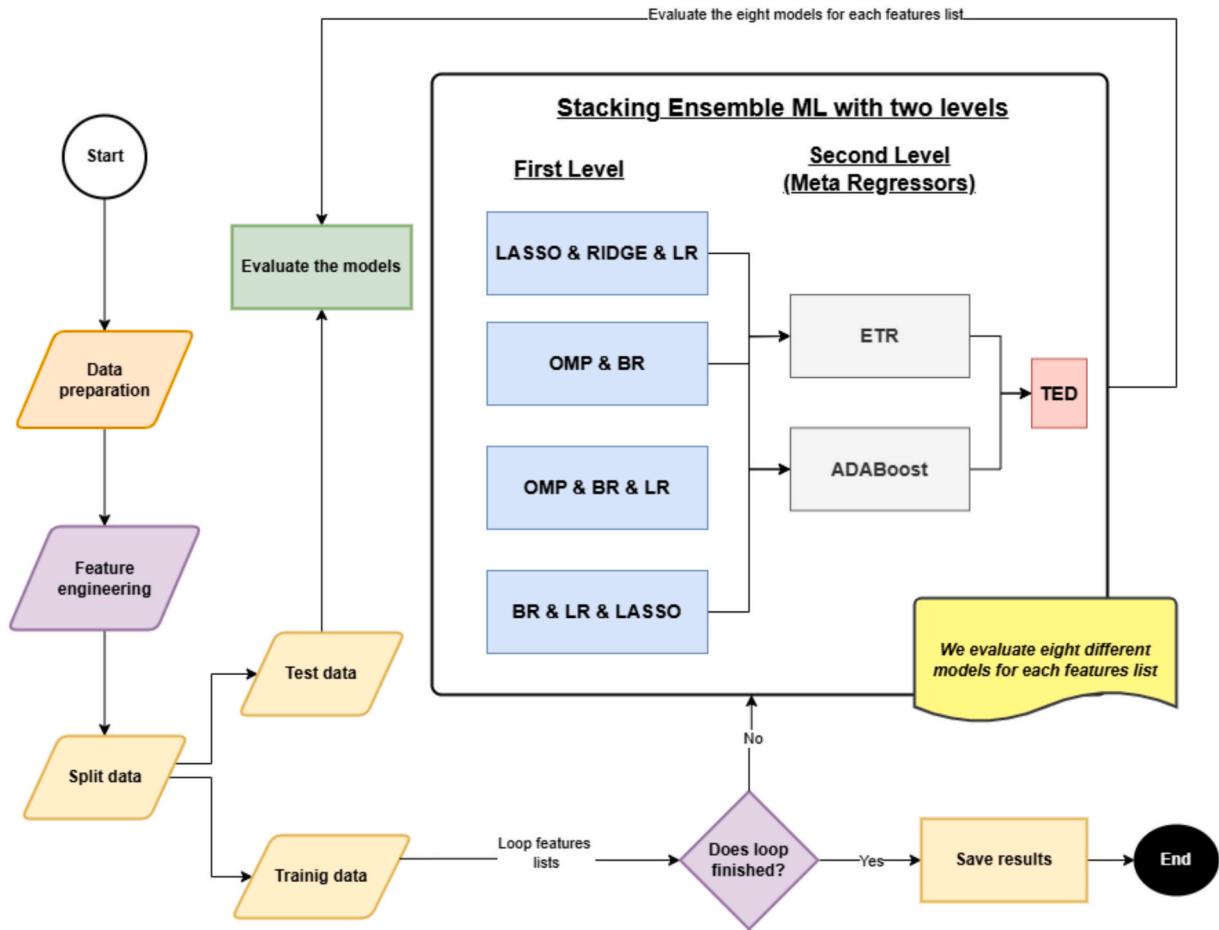


Fig. 2. Proposed SEML framework for predicting TED using a two-level stacking ensemble.

combination of a few bases from a predetermined set of base functions. Bases are chosen in a greedy manner by adding the base with the highest correlation to the current approximation's residual error. This process continues until a stopping criterion is reached, such as a maximum number of bases or a minimum error threshold. OMP offers various advantages, including computational efficiency with a complexity that scales linearly with signal length. It also aids in feature selection and dimensionality reduction by automatically selecting the most relevant bases from the dictionary. Users need to specify the dictionary of base functions and the stopping condition. The algorithm then updates the signal approximation and iteratively selects atoms from the dictionary until the stopping requirement is met.

Ridge Regression [63] is a type of linear regression algorithm used in machine learning to predict a target variable. It incorporates a regularization factor into the cost function to prevent overfitting and improve the model's ability to generalize. The main goal of ridge regression is to find the coefficients that minimize the sum of the squared residuals between the predicted and actual values of the target variable, while also restricting the sum of the squared coefficients. By limiting the coefficient magnitude, this regularization term prevents overfitting and encourages the model to use a smaller, more concise set of features.

Linear Regression (LR) [61] is a statistical method used to model the relationship between a dependent variable and one or more independent variables. It is commonly employed to predict a continuous numerical output based on a set of input variables. The fundamental concept behind linear regression is to find the coefficients for a linear equation that can be used to predict the target variable based on the input variables. This is achieved by utilizing an optimization algorithm, such as gradient descent, to iteratively adjust the coefficients in order to

minimize the difference between the predicted values and the actual values in the training data. By doing so, the model aims to provide the best fit to the data and make accurate predictions for new data points.

Lasso Regression [62] is short for Least Absolute Shrinkage and Selection Operator, is a type of linear regression method used for feature selection and regularization. It involves using a set of input variables to predict a continuous numerical outcome. The primary objective of lasso regression is to minimize the sum of the squared differences between the predicted and actual values of the target variable, while also adhering to a constraint on the sum of their absolute values. This constraint effectively shrinks the less important feature coefficients to zero, thereby performing feature selection as well. By controlling the magnitude of the coefficients, lasso regression helps to prevent overfitting and can be particularly useful when dealing with high-dimensional datasets where feature selection is crucial for model interpretability and performance.

Bayesian Ridge Regression (BR) [61] is a model that uses Bayesian statistics to incorporate prior information and uncertainty. Its main goal is to find the coefficients that maximize the posterior probability of the model, given the data and prior knowledge. This involves calculating the posterior distribution of the coefficients using Bayes' theorem and using an optimization technique to find the values that maximize this distribution. One key advantage of Bayesian Ridge Regression in machine learning is its ability to include previous information and uncertainty in the model, potentially improving regularization and generalization. Before using Bayesian Ridge Regression in a machine learning task, users need to specify the target variable and the input feature set for prediction. They also need to specify the prior distribution of the coefficients and the regularization parameter, which controls the strength of the regularization. The model then calculates the posterior

distribution of the coefficients using Bayes' theorem and an optimization method, and finds the values that maximize this distribution.

4.4. Applying ANN approach

The second approach uses artificial neural networks (ANNs) algorithm [66], also known as neural networks or neural nets, which are machine learning models that are constructed based on the principles of neuronal organization found in biological neural networks that make up animal brains. The ANN comprises an input layer, an output layer, and one or more hidden layers. For our proposed framework, we will have one hidden layer as shown in Fig. 3.

4.5. Evaluation matrix

In our analysis of transportation energy prediction, we assessed the effectiveness of the machine learning algorithm by utilizing five regression metrics: R-Squared (R^2), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). These metrics enabled us to evaluate the precision and performance of our predictions.

For the following equations consider:

- (i) is the iteration number
- (n) is the number of data points
- (Y_i) is the actual value.
- ($\hat{Y}_i$) is the predicted value.
- (Y^-) is the mean of the actual values.

R^2 , also known as the coefficient of determination, is a metric used to

evaluate how well a model fits the data. It represents the proportion of variation in the dependent variable that can be predicted from the independent variable(s). R^2 values range from 0 to 1, with higher values indicating a better fit.

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - Y^-)^2} \quad (1)$$

1. R-squared (R^2)

MSE is a metric which measures the average of the squares of the errors, which is the average squared difference between estimated values and actual values. It quantifies how close a model's predictions are to the true values, with larger errors being penalized more because they are squared. A lower MSE indicates better model performance. However, MSE is sensitive to outliers due to the squaring of errors, so it's important to consider the context and distribution of the data when using this metric.

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \quad (2)$$

2. Mean squared error (MSE)

RMSE, is one of commonly used measures that compare predicted values to observed values. It is calculated as the square root of the MSE, providing a standard deviation of the prediction errors. Since RMSE is expressed in the same units as the target variable, it offers a clearer interpretation than MSE. A lower RMSE indicates more accurate pre-

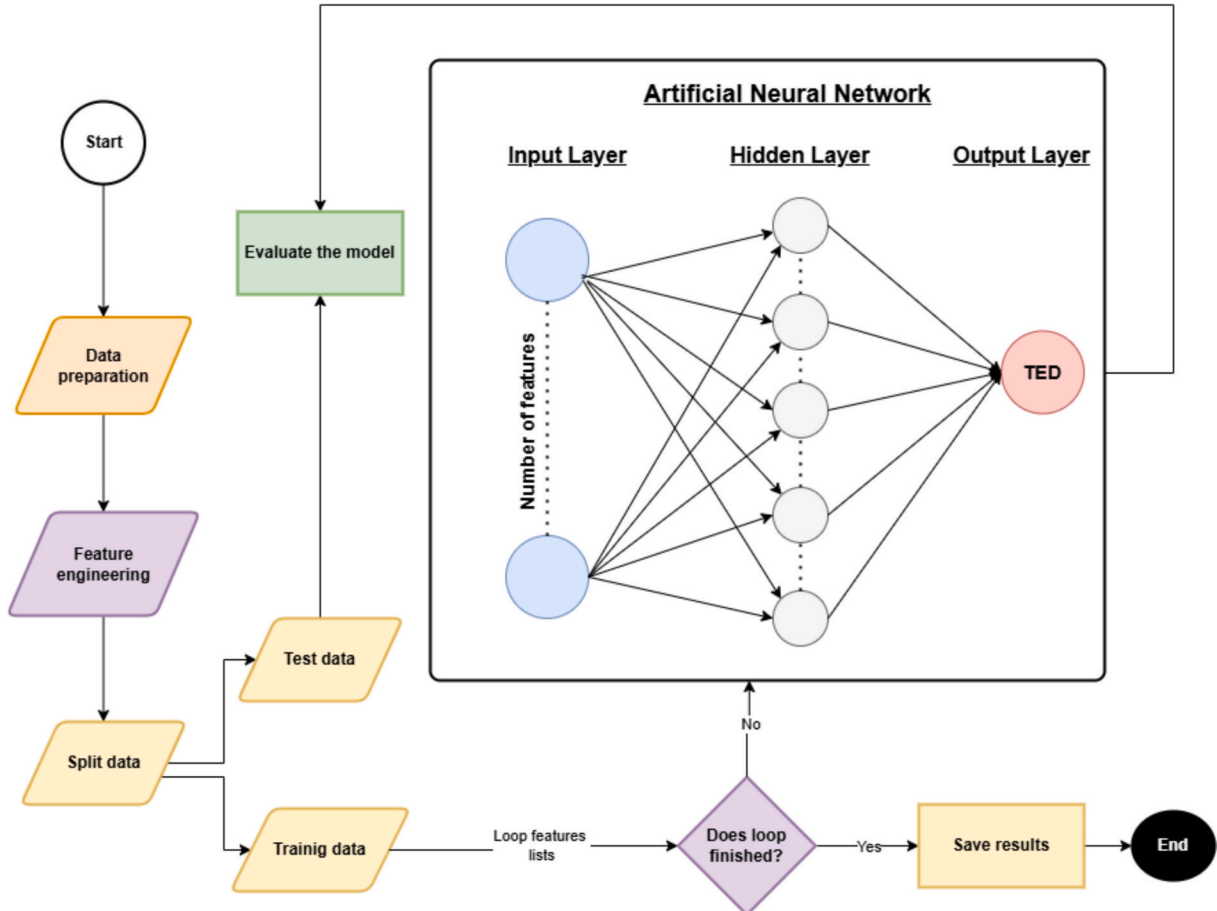


Fig. 3. Proposed framework for predicting TED using ANN.

dictions, while a higher RMSE suggests greater discrepancies between predicted and actual values.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2} \quad (3)$$

3. Root Mean square error (RMSE)

MAE is a regression metric that quantifies the average size of errors between predicted and actual values, regardless of their direction. It is calculated by taking the average of the absolute differences between predictions and actual observations. MAE treats all errors equally, meaning that it does not penalize larger errors more than smaller ones. A lower MAE value signifies better performance of the model.

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i| \quad (4)$$

4. Mean absolute error (MAE)

MAPE is a metric that measures the accuracy of a regression model as a percentage. It calculates the average absolute difference between the actual and predicted values, expressed as a percentage of the actual values. This makes it easy to interpret, as it presents error in percentage terms, which is useful for comparing performance across different datasets. However, MAPE can be unreliable when actual values are close to zero, as this can result in very large or undefined percentage errors.

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{Y_i - \hat{Y}_i}{Y_i} \right| \quad (5)$$

5. Mean absolute percentage error (MAPE)

5. Results

We conducted a comprehensive analysis to explore the intricate relationships among the distinct categories of factors: country-based factors, which include transport and socioeconomic elements, as well as global influences. We first evaluated each category individually to gain insights into their unique characteristics and performance as shown in Table 3 before investigating their interconnections.

Following this, we examined how the features of these categories combine and impact the results. This led us to identify the most significant features for predicting TED as presented in Table 4 which summarize the category of features and is used effectively along with Table 3 to address our first research question (RQ1). Additionally, we developed a generalized model to predict TED, which answers our second research question (RQ2).

In the following sections, we will begin by:

1. Performing traditional regression analysis on our dataset.
2. Introducing our analysis for each category individually.
3. Assess the importance of the used features.

4. Presenting the results of our top five models for predicting TED.
5. Analyzing predictions for TED in Turkey for the year 2040.

5.1. Baseline model performance without stacking

Prior to presenting our significant findings, we conducted an analysis of traditional regression models to underscore the advantages of utilizing stacking models, which enhance the performance of weaker learners. As illustrated in Table 2, we assessed the predictive performance of baseline traditional regression and ensemble models, including ETR, BR, LR, Ridge, AdaBoost, LASSO, and OMP, without the application of stacking. Among these models, ETR and BR achieved the most robust baseline results, yielding the lowest error values (MSE $\approx$ 6.4614, RMSE $\approx$ 2.5419 and MAE $\approx$ 1.8340) and demonstrating a high goodness of fit ($R^2 \approx$ 0.9850). This suggests their capacity to partially capture the nonlinear patterns present in transportation energy demand. Conversely, LR and Ridge exhibited slightly lower accuracy, reflecting the constraints of linear assumptions when modeling complex and heterogeneous transportation systems. Sparse regularization-based models, such as LASSO and OMP, demonstrated considerably weaker performance, with higher error metrics and diminished R^2 values, indicating underfitting due to aggressive coefficient shrinkage. Although AdaBoost maintained a relatively high R^2 , it suffered from a significantly elevated MAPE, revealing its instability as a standalone learner and its sensitivity to data distribution. Overall, while these non-stacked models provide reasonable baseline performance, their predictive accuracy consistently falls short compared to the proposed stacking ensemble approach. This observation highlights the advantages of stacking, which integrates complementary learning behaviors from multiple base models. Such integration results in improved robustness and generalization across diverse patterns of transportation energy demand. Consequently, this motivates the adoption of the SEML framework, whose quantitative performance gains over these baselines are detailed and analyzed in the next sections.

5.2. Features result by category

As shown in Table 3 our findings from the transport category proved particularly noteworthy. The best result among the other algorithms achieved using the SEML of OMP, and BR at the first level, with the ETR serving as the meta regressor among the other algorithms with an impressive R^2 value of 0.9944, underscoring the model's strong predictive capability. Upon excluding the investment-related features (INV-RD and INV-RL) from our analysis, we obtained a MAPE of 15.5393. Furthermore, we calculated a MSE of 2.4273, RMSE of 1.5580, and a MAE of 0.9550. However, upon reintroducing the investment features into the model, we observed a slight decline in the R^2 value to 0.9938. Notably, this revised model exhibited an increase in the MAPE to 27.4079, while the MSE rose to 2.6794, the RMSE adjusted to 1.6369, and the MAE increased to 1.1425.

As we shifted our focus to the socioeconomic category features only, the results yield an R^2 value of 0.9931. In this analysis, we recorded a

Table 2
Evaluation matrix for base models against all features.

Algorithm	Train					Test				
	MAPE	R^2	MSE	RMSE	MAE	MAPE	R^2	MSE	RMSE	MAE
ETR	28.5119	0.9771	10.7323	3.2760	2.1196	33.6678	0.9850	6.4614	2.5419	1.8340
BR	28.5119	0.9771	10.7323	3.2760	2.1196	33.6678	0.9850	6.4614	2.5419	1.8340
LR	29.0280	0.9772	10.7075	3.2722	2.1050	34.3484	0.9847	6.5810	2.5654	1.8367
RIDGE	35.6278	0.9680	15.0019	3.8732	2.5877	42.1423	0.9763	10.2131	3.1958	2.3371
AdaBoost	106.6392	0.9760	11.2679	3.3568	2.5943	111.4628	0.9690	13.3286	3.6508	2.6763
LASSO	174.7332	0.8960	48.7328	6.9809	5.2302	178.5100	0.9016	42.3345	6.5065	5.0409
OMP	152.8911	0.8631	64.1471	8.0092	5.5363	156.0336	0.8823	50.6377	7.1160	5.2225

Table 3

Features results by category using SEMML of OMP, and BR at first level and ETR as meta regressor.

Category	Number of features	Train					Test				
		MAPE	R ²	MSE	RMSE	MAE	MAPE	R ²	MSE	RMSE	MAE
SEML Transport without investment features	4	11.2112	0.9938	2.9283	1.7112	0.8980	15.5393	0.9944	2.4273	1.5580	0.9550
Transport	6	10.9032	0.9954	2.1508	1.4666	0.8409	27.4079	0.9938	2.6794	1.6369	1.1425
Socioeconomic	2	32.5606	0.9899	4.7484	2.1791	1.1441	46.6229	0.9931	2.9801	1.7263	1.1019
Socioeconomic, and Global	3	62.1903	0.9852	6.9395	2.6343	1.4046	71.1295	0.9918	3.5393	1.8813	1.2653

Table 4

List of best models features.

Model	Features Category	Number of Features	Features
Model-A	Transport (6)	9	TON-KM-RD, TON-KM-RL, P-KM-RD, P-KM-RL, INV-RD, INV-RL, GDP, POP
Model-B	Socioeconomic (2)	7	OIL-AP
	Global (1)		TON-KM-RD, TON-KM-RL, P-KM-RD, P-KM-RL
	Transport (4)		GDP, POP
Model-C	Socioeconomic (2)	6	OIL-AP
	Global (1)		TON-KM-RD, TON-KM-RL, P-KM-RD, P-KM-RL
	Transport (4)		GDP, POP
Model-D	Socioeconomic (2)	5	.
	Global (0)		TON-KM-RD, TON-KM-RL, P-KM-RD, P-KM-RL
	Transport (4)		.
Model-E	Socioeconomic (0)	3	OIL-AP
	Global (1)		P-KM-RD
	Transport (1)		GDP, POP
	Socioeconomic (2)		.
	Global (0)		.

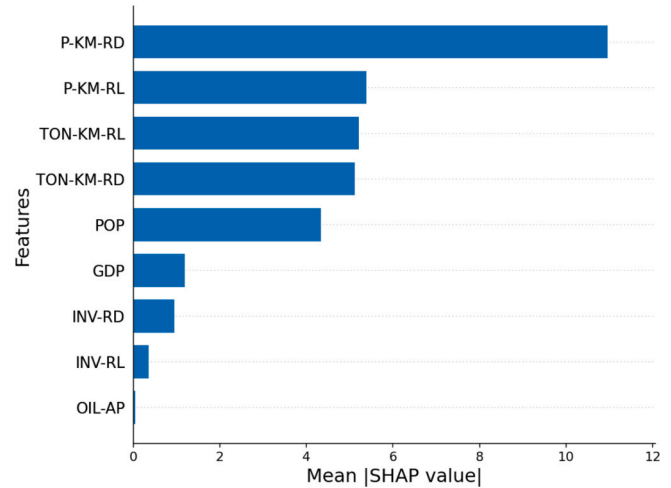
MAPE of 46.6229, an MSE of 2.9801, an RMSE of 1.7263, and an MAE of 1.1019, all obtained using the same advanced SEMML technique.

Examining the global category alone is not worthwhile since it consists of only one feature. Furthermore, we analyzed the combination of socioeconomic and global factors demonstrated a R² value of 0.9918, and produced a higher MAPE of 71.1295, MSE of 3.5393, RMSE of 1.8813, and MAE of 1.2653 by the same SEMML model.

On the other hand, the combination of transport and socioeconomic factors, along with transport and global factors, significantly improved predictive accuracy. This emphasizes their importance when analyzed together, which we will discuss further in the next section.

5.3. Feature importance analysis

To assess the relative importance of the selected features and enhance the interpretability of our proposed framework, a SHAP (SHapley Additive exPlanations) [67] based feature importance analysis was developed. The resulting mean absolute SHAP values, aggregated across all countries, reveal that transportation activity variables are the dominant drivers of TED prediction. Passenger-kilometers by road (P-KM-RD) exhibit the highest influence, followed by passenger and freight related rail and road indicators (P-KM-RL, TON-KM-RL, and TON-KM-RD). Population shows a moderate contribution, whereas macroeconomic and infrastructure-related variables, including GDP and investment indicators, have comparatively lower impact on the model output, as summarized in Fig. 4. Although the average oil price variable (OIL-AP) presents a near-zero mean SHAP value when considered independently, its effect on energy demand prediction emerges through interactions with other predictors particularly transport activity variables

**Fig. 4.** Feature importance results using SHAP method.

indicating an indirect but meaningful role in shaping demand dynamics.

5.4. Results of applying ML techniques

In this section we explain the results of the top five models against the SEMML, and ANN algorithms which mentioned in Table 4 starting from Model-A which represents all the nine features including the transportation, socioeconomic, and global categories as shown in Table 5 the SEMML of OMP, and BR in the first level and ETR in the second level has the best results with R² 0.9961, and values 9.8489, 1.6615, 1.2890, 0.7512 for MAPE, MSE, RMSE, MAE respectively. The SEMML of LASSO, RIDGE, and LR in the first level and ETR in the second level came in the second rank with R² 0.9955, and values 10.6787, 1.9286, 1.3887, 0.7781 for MAPE, MSE, RMSE, MAE respectively. The ETR in the second level for the other algorithms [OMP, BR, LR], as well as [BR, LR, LASSO], still yield results with R² 0.9946, and 0.9944 respectively. The ANN yields R² 0.9867, while the ADABOOST as a meta regressor has R² results ranging between 0.9484 and 0.9541.

Table 6 presents the results for Model-B, which includes seven features encompassing all categories except transport investment. The best-performing model was the SEMML configuration using LASSO, Ridge, and LR at the first level, and ETR as the meta-regressor at the second level. This model achieved the highest performance with an R² value of 0.9956, and error metrics of 17.1219 (MAPE), 1.9050 (MSE), 1.3802 (RMSE), and 0.8237 (MAE). The second-best performance was achieved by the SEMML model using OMP and BR in the first level, and ETR as the meta-regressor. This configuration obtained an R² of 0.9949, with error values of 15.2414 (MAPE), 2.2086 (MSE), 1.4861 (RMSE), and 0.7653 (MAE). Additionally, the ETR meta-regressor achieved an R² of 0.9937 when paired with [OMP, BR, LR], and 0.9909 with [BR, LR, LASSO]. The ANN model recorded an R² of 0.9841. Meanwhile, the AdaBoost meta-regressor produced R² values ranging between 0.9523 and 0.9587 across different base model configurations.

Table 7 shows the results with Model-C that consists of six features

Table 5
Evaluation matrix for Model-A.

Algorithm		Train					Test				
		MAPE	R ²	MSE	RMSE	MAE	MAPE	R ²	MSE	RMSE	MAE
SEML	OMP_BR_ETR	5.6690	0.9932	3.1669	1.7796	0.7381	9.8489	0.9961	1.6615	1.2890	0.7512
	LASSO_RIDGE_LR_ETR	4.9956	0.9952	2.2577	1.5026	0.6117	10.6787	0.9955	1.9286	1.3887	0.7781
	OMP_BR_LR_ETR	6.0273	0.9933	3.1406	1.7722	0.8109	11.0329	0.9946	2.3235	1.5243	0.8869
	BR_LR_LASSO_ETR	5.2189	0.9930	3.2615	1.8060	0.7579	10.2383	0.9944	2.3911	1.5463	0.8690
	ANN	22.2517	0.9946	2.5505	1.5970	0.9681	21.9570	0.9867	5.3447	2.3119	1.6809
SEML	LASSO_RIDGE_LR_ADABOOST	100.7915	0.9601	18.6893	4.3231	3.0976	105.9750	0.9541	19.7468	4.4437	3.0924
	BR_LR_LASSO_ADABOOST	109.5451	0.9674	15.2832	3.9094	2.7930	115.4965	0.9539	19.8422	4.4545	3.0771
	OMP_BR_LR_ADABOOST	100.3052	0.9569	20.1958	4.4940	3.1364	106.5682	0.9484	22.2000	4.7117	3.2603
	OMP_BR_ADABOOST	100.2985	0.9570	20.1656	4.4906	3.1354	106.5682	0.9484	22.2000	4.7117	3.2603

Table 6
Evaluation matrix for Model-B.

Algorithm		Train					Test				
		MAPE	R ²	MSE	RMSE	MAE	MAPE	R ²	MSE	RMSE	MAE
SEML	LASSO_RIDGE_LR_ETR	9.3682	0.9957	2.0193	1.4210	0.6655	17.1219	0.9956	1.9050	1.3802	0.8237
	OMP_BR_ETR	8.8318	0.9929	3.3290	1.8246	0.8234	15.2414	0.9949	2.2086	1.4861	0.7653
	OMP_BR_LR_ETR	9.3499	0.9944	2.6382	1.6243	0.7944	18.4764	0.9937	2.6889	1.6398	0.8712
	BR_LR_LASSO_ETR	10.8774	0.9937	2.9418	1.7152	0.8224	20.8849	0.9909	3.8975	1.9742	1.0333
	ANN	15.6800	0.9935	3.0784	1.7545	1.1365	13.3518	0.9841	6.3726	2.5244	1.5639
SEML	LASSO_RIDGE_LR_ADABOOST	115.2030	0.9549	21.1310	4.5968	3.3564	119.1500	0.9587	17.7530	4.2134	3.0110
	OMP_BR_LR_ADABOOST	110.7768	0.9589	19.2718	4.3900	3.1595	115.6735	0.9523	20.5013	4.5278	3.0967
	OMP_BR_ADABOOST	100.5477	0.9546	21.2597	4.6108	3.2563	105.9389	0.9523	20.5085	4.5286	3.2170
	BR_LR_LASSO_ADABOOST	110.0398	0.9519	22.5291	4.7465	3.4210	114.8108	0.9523	20.5363	4.5317	3.2171

Table 7
Evaluation matrix for Model-C.

Algorithm		Train					Test				
		MAPE	R ²	MSE	RMSE	MAE	MAPE	R ²	MSE	RMSE	MAE
SEML	LASSO_RIDGE_LR_ETR	8.3123	0.9961	1.8077	1.3445	0.6528	14.6717	0.9963	1.5728	1.2541	0.7727
	OMP_BR_ETR	6.0594	0.9935	3.0286	1.7403	0.7893	11.7109	0.9951	2.1007	1.4494	0.8170
	OMP_BR_LR_ETR	5.8371	0.9945	2.5897	1.6092	0.7873	13.9312	0.9937	2.7242	1.6505	0.9208
	BR_LR_LASSO_ETR	8.8641	0.9929	3.3120	1.8199	0.8202	18.7236	0.9916	3.6155	1.9014	1.0714
	ANN	18.7167	0.9875	5.8906	2.4270	1.5565	14.4106	0.9850	6.0303	2.4557	1.5343
SEML	OMP_BR_ADABOOST	99.5955	0.9575	19.9179	4.4629	3.1156	104.9969	0.9600	17.2219	4.1499	2.9767
	BR_LR_LASSO_ADABOOST	99.3888	0.9649	16.4561	4.0566	2.7641	105.4923	0.9600	17.2260	4.1504	2.8702
	OMP_BR_LR_ADABOOST	100.6063	0.9542	21.4490	4.6313	3.2464	106.0879	0.9580	18.0500	4.2485	3.1160
	LASSO_RIDGE_LR_ADABOOST	103.5321	0.9541	21.4970	4.6365	3.2914	109.0002	0.9546	19.5141	4.4175	3.1996

including the transportation category without investment features, and the socioeconomic category. The SEML of LASSO, RIDGE, and LR in the first level and ETR in the second level came in the first rank with R² 0.9963, and values 14.6717, 1.5728, 1.2541, 0.7727 for MAPE, MSE, RMSE, MAE respectively. The SEML OMP, and BR in the first level, and ETR as the meta regressor in the second rank with R² 0.9951, and values 11.7109, 2.1007, 1.4494, 0.8170 for MAPE, MSE, RMSE, MAE respectively. The ETR meta regressor achieved R² scores of 0.9937 and 0.9916 with algorithms [OMP, BR, LR], and [BR, LR, LASSO] respectively. The

ANN also achieved an R² score of 0.9850. The remaining algorithms paired with the ADABOOST meta regressor achieved R² scores ranging between 0.9546 and 0.9600.

Table 8 presents the results for Model-D, which consists of five features drawn from the transport category (excluding investment features) and the global category. The best-performing configuration was the SEML model using OMP and BR at the first level, and ETR as the meta-regressor at the second level. This model achieved an R² of 0.9951, with error metrics of 14.0124 (MAPE), 2.1276 (MSE), 1.4586 (RMSE), and

Table 8
Evaluation matrix for Model-D.

Algorithm		Train					Test				
		MAPE	R ²	MSE	RMSE	MAE	MAPE	R ²	MSE	RMSE	MAE
SEML	OMP_BR_ETR	11.1485	0.9935	3.0277	1.7400	0.9180	14.0124	0.9951	2.1276	1.4586	0.8548
	OMP_BR_LR_ETR	11.1449	0.9941	2.7481	1.6578	0.8971	13.9658	0.9941	2.5345	1.5920	0.9348
	LASSO_RIDGE_LR_ETR	21.8935	0.9898	4.7619	2.1822	1.2050	28.8180	0.9810	8.1789	2.8599	1.6985
	BR_LR_LASSO_ETR	19.4421	0.9868	6.1788	2.4857	1.3445	18.8460	0.9808	8.2677	2.8754	1.7235
	ANN	25.3176	0.9919	3.8343	1.9581	1.1618	12.1559	0.9806	7.8095	2.7946	1.8446
SEML	LASSO_RIDGE_LR_AdaBoost	111.2549	0.9623	17.6860	4.2055	3.1942	114.8133	0.9632	15.8195	3.9774	3.0672
	BR_LR_LASSO_AdaBoost	111.2144	0.9608	18.3769	4.2868	3.2111	115.7049	0.9597	17.3520	4.1656	3.1248
	OMP_BR_LR_AdaBoost	149.5928	0.9618	17.8916	4.2298	3.3623	155.2993	0.9459	23.2582	4.8227	3.5659
	OMP_BR_AdaBoost	149.5928	0.9618	17.8916	4.2298	3.3623	155.2993	0.9459	23.2582	4.8227	3.5659

0.8548 (MAE). The second-best configuration was the SEML model combining OMP, BR, and LR in the first level and ETR in the second level. It achieved an R^2 of 0.9941, with corresponding error values of 13.9658 (MAPE), 2.5345 (MSE), 1.5920 (RMSE), and 0.9348 (MAE). Other SEML configurations, including [LASSO, Ridge, LR] and [BR, LR, LASSO] with ETR as the *meta*-regressor, along with the ANN model, yielded slightly lower R^2 scores of 0.9810, 0.9808, and 0.9806, respectively. Models using AdaBoost as the *meta*-regressor resulted in R^2 values ranging from 0.9459 to 0.9632 across different base learner combinations.

Also, the study provides a significant result shown in Table 9 for Model-E which contains only three features, including only one feature from the transport category, and the socioeconomic category. The SEML of LASSO, RIDGE, and LR in the first level and ETR in the second level came in the first rank with R^2 0.9952, and values 7.7656, 2.0542, 1.4332, 0.9092 for MAPE, MSE, RMSE, MAE respectively. The SEML BR, LR, and LASSO in the first level with ETR as the meta regressor in the second rank with R^2 0.9946, and values 7.1932, 2.3255, 1.5249, 0.9000 for MAPE, MSE, RMSE, MAE respectively. The algorithms [OMP, BR, LR], along with [OMP, BR] with ETR as a meta regressor, achieved R^2 scores of 0.9939 and 0.9934, respectively. The other results, including the algorithms with ADABOOST as a meta regressor and ANN, achieved R^2 scores ranging from 0.9332 to 0.9459.

After examining the best five models over the nine algorithms, the results of these experiments showed that the SEML which consists of LASSO, RIDGE, and LR in the first level and ETR in the second level as the meta regressor for Model-C which consists of six features containing the transport features without transportation investment parameters and the socioeconomic features has the best results with R^2 0.9963, and values 9.8489, 1.6615, 1.2890, 0.7512 for MAPE, MSE, RMSE, MAE respectively. The SEML model which consists of OMP, and BR in the first level and ETR in the second level as the meta regressor has the best results across all models as shown in Fig. 5 which represents the average R^2 of the nine algorithms against the five models. The performance is truly remarkable for both algorithms, they produce excellent results across all models, showing a significant R^2 value, as presented in Fig. 6.

To validate our model, we utilized data from Turkey, which was intentionally excluded during the training phase to serve as an unseen test set. The objective was to forecast TED for the final three years of the dataset, specifically from 2018 to 2020. The model exhibited strong predictive performance, achieving an accuracy exceeding 95%. The predicted TED values for the years 2020, 2019, and 2018 were 25.94, 28.02, and 27.85 MTOE, respectively, compared to the actual recorded values of 26.89, 26.60, and 26.50 MTOE.

5.5. Turkey TED prediction for 2040

In the field of transportation planning, accurate projections of energy demand play a vital role in shaping future strategies. A recent study by Korkmaz [29] focused on forecasting energy demand in Turkey under two distinct scenarios. These scenarios were developed using the least squares method to estimate model parameters. In the first scenario, the

current distribution of demand across various transportation modes is maintained, and TED is projected to 2040 based on existing trends. In contrast, the second scenario assumes a 5% modal shift from highway transportation to other modes that have gained favor due to increased flexibility and efficiency in recent years. This scenario highlights the potential impact of growing demand for alternative transportation modes on TED particularly significant in Turkey, where approximately 90% of travel demand is currently met via highways.

The least squares method [68], employed in the study, is a widely used technique for estimating unknown parameters in a model. It minimizes the sum of the squared differences between observed values and the model's predicted outputs [69]. According to Korkmaz [29], the projected TED values for 2040 are 63.58 MTOE under Scenario 1 and 41.56 MTOE under Scenario 2. In comparison, our study predicts TED values of 41.05 MTOE and 40.24 MTOE for the same scenarios, respectively, as shown in Fig. 7. We believe the primary distinction between our work and that of Korkmaz lies in the inclusion of two additional transportation modes: air and water. This inclusion likely accounts for the significantly smaller difference between the two scenarios in our results approximately 1 MTOE compared to the 22 MTOE difference observed in Korkmaz's projections.

In conclusion, both studies indicate that a modal shift away from highway transportation toward alternatives such as rail, water, and air is expected to contribute to a reduction in Turkey's TED by 2040.

6. Discussion

6.1. Models' results discussion

As illustrated in Fig. 8, which presents the R^2 results of the algorithms across different five models ordered by the average R^2 for the five models, all SEML algorithms utilizing ETR as the meta regressor in the second level have consistently demonstrated exceptional predictive performance in estimating TED across multiple countries in our study.

The highest-performing configuration was the SEML model composed of OMP and BR at the first level, with ETR at the second level. It achieved R^2 scores of 0.9961, 0.9949, 0.9951, 0.9951, and 0.9934 for Models A through E, respectively. Ranking second was the SEML model incorporating OMP, BR, and LR at the first level, also paired with ETR, which achieved R^2 scores of 0.9946, 0.9937, 0.9937, 0.9941, and 0.9939.

In third place, the SEML configuration combining LASSO, Ridge, and LR in the first level again with ETR as the *meta*-regressor achieved R^2 scores of 0.9955, 0.9956, 0.9963, 0.9810, and 0.9952. Fourth place was occupied by the SEML model using BR, LR, and LASSO at the first level, also with ETR, yielding R^2 scores of 0.9944, 0.9909, 0.9916, 0.9808, and 0.9946.

The ANN ranked fifth, with R^2 values of 0.9867, 0.9841, 0.9850, 0.9806, and 0.9358 across the five models. In contrast, SEML configurations using AdaBoost as the *meta*-regressor performed less favorably. The best result for AdaBoost was an R^2 of 0.9632, achieved by the SEML configuration of LASSO, Ridge, and LR in Model-D. Based on these

Table 9
Evaluation matrix for Model-E.

Algorithm		Train					Test				
		MAPE	R^2	MSE	RMSE	MAE	MAPE	R^2	MSE	RMSE	MAE
SEML	LASSO_RIDGE_LR_ETR	6.3730	0.9876	5.8315	2.4148	0.8631	7.7656	0.9952	2.0542	1.4332	0.9092
	BR_LR_LASSO_ETR	7.3357	0.9863	6.4234	2.5344	0.9275	7.1932	0.9946	2.3255	1.5249	0.9000
	OMP_BR_LR_ETR	7.8243	0.9763	11.1233	3.3352	1.3854	6.2998	0.9939	2.6385	1.6243	0.9492
	OMP_BR_ETR	8.7966	0.9761	11.2208	3.3498	1.3765	7.7179	0.9934	2.8359	1.6840	0.9878
	OMP_BR_LR_ADABOOST	100.7907	0.9337	31.0562	5.5728	3.4972	105.1692	0.9459	23.2588	4.8227	3.4043
	OMP_BR_ADABOOST	100.7907	0.9337	31.0562	5.5728	3.4972	105.1692	0.9459	23.2588	4.8227	3.4043
	LASSO_RIDGE_LR_ADABOOST	100.8340	0.9450	25.7959	5.0790	3.5246	106.1464	0.9411	25.3413	5.0340	3.5667
	ANN	38.1057	0.9619	17.9709	4.2392	2.7954	21.0342	0.9358	25.8082	5.0802	3.3976
SEML	BR_LR_LASSO_ADABOOST	101.7324	0.9261	34.6388	5.8855	3.8308	106.7990	0.9332	28.7322	5.3602	3.8161

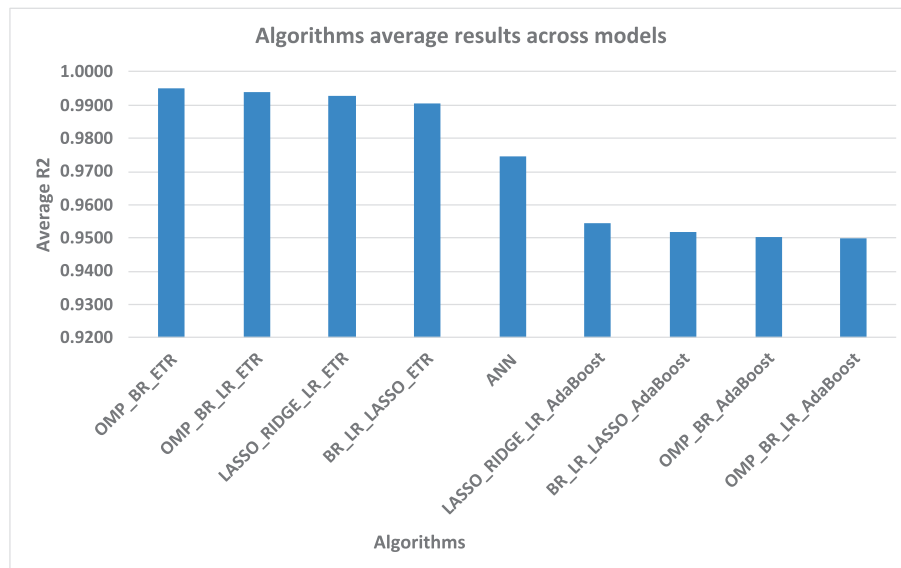


Fig. 5. The evaluation of all the SEML and the ANN frameworks using the average R^2 value of each algorithm across the five models.

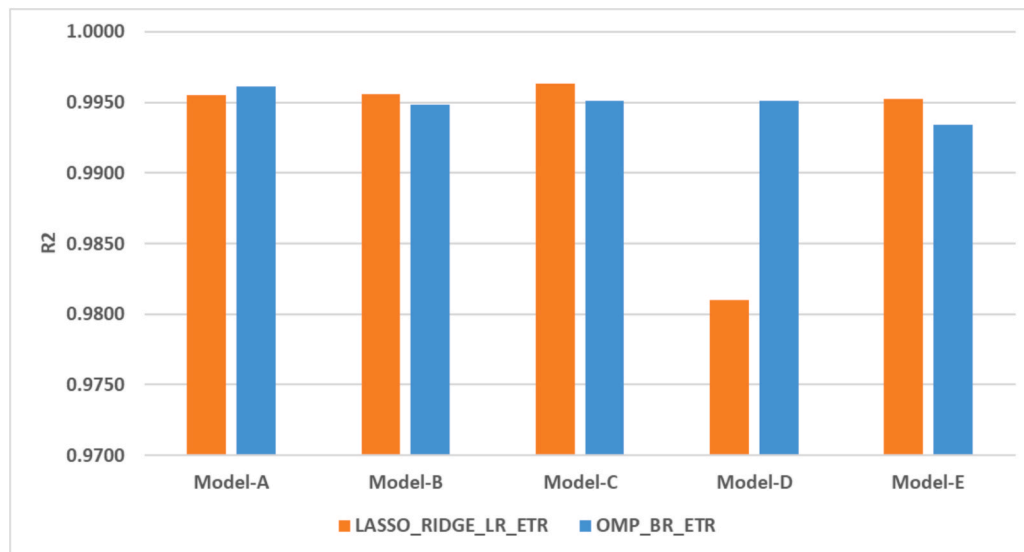


Fig. 6. The evaluation of the SEML consists of [LASSO, RIDGE, LR], and [OMP, BR] in the first level and ETR in the second level as the meta regressor across the five models.

findings, we do not recommend using AdaBoost as the *meta*-regressor; rather, we strongly advocate for the use of ETR due to its consistent and superior performance.

Among all models, Model-C, which includes six features combining the transport category (excluding investment-related features) and the socioeconomic category, produced the best results when using the SEML algorithm of LASSO, Ridge, and LR at the first level and ETR at the second level. Notably, Model-E which includes only three features: one transport-related (passenger kilometers for road) and two socioeconomic features (GDP and population) also achieved impressive results. This highlights the feasibility of accurately predicting TED even in countries with limited transportation infrastructure or sparse data availability, provided that key socioeconomic indicators are included.

Interestingly, our results revealed that investment-related features within the transport category (INV-RD and INV-RL) had minimal impact on the overall model performance, suggesting they are not significant contributors to TED prediction in the context of our study.

6.2. Proposed framework accuracy cross trained countries

The country-level evaluation, with accuracy calculated over the dataset for each nation, demonstrates that the proposed framework delivers exceptionally strong and consistent performance across diverse regions and transportation contexts. The results are obtained using the proposed SEML algorithm, which combines LASSO, Ridge, and LR at the first level, and ETR at the second level through Model-C. Among European countries, France achieves the highest accuracy at 99.13%, followed by the United Kingdom (98.89%), Spain (97.11%), Finland (97.15%), Sweden (96.51%), the Czech Republic (96.72%), Switzerland (96.02%), Poland (94.30%), Italy (92.53%), Hungary (93.19%), and Norway (89.25%). Overall, the average accuracy for European countries exceeds 95.53%, confirming the model's ability to capture the fundamental factors driving transportation energy demand despite differences in modal splits and economic conditions. Beyond Europe, the model maintains excellent generalization, achieving 98.41% in Australia, 96.49% in Japan, and 93.13% in Russia, demonstrating its robustness

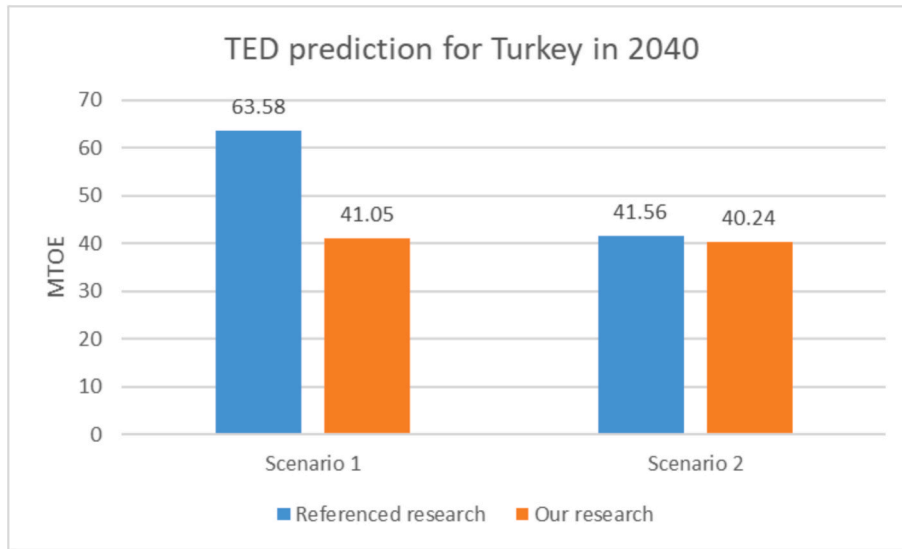


Fig. 7. TED prediction for Turkey in 2040 in compare with the research conducted by Korkmaz [29].

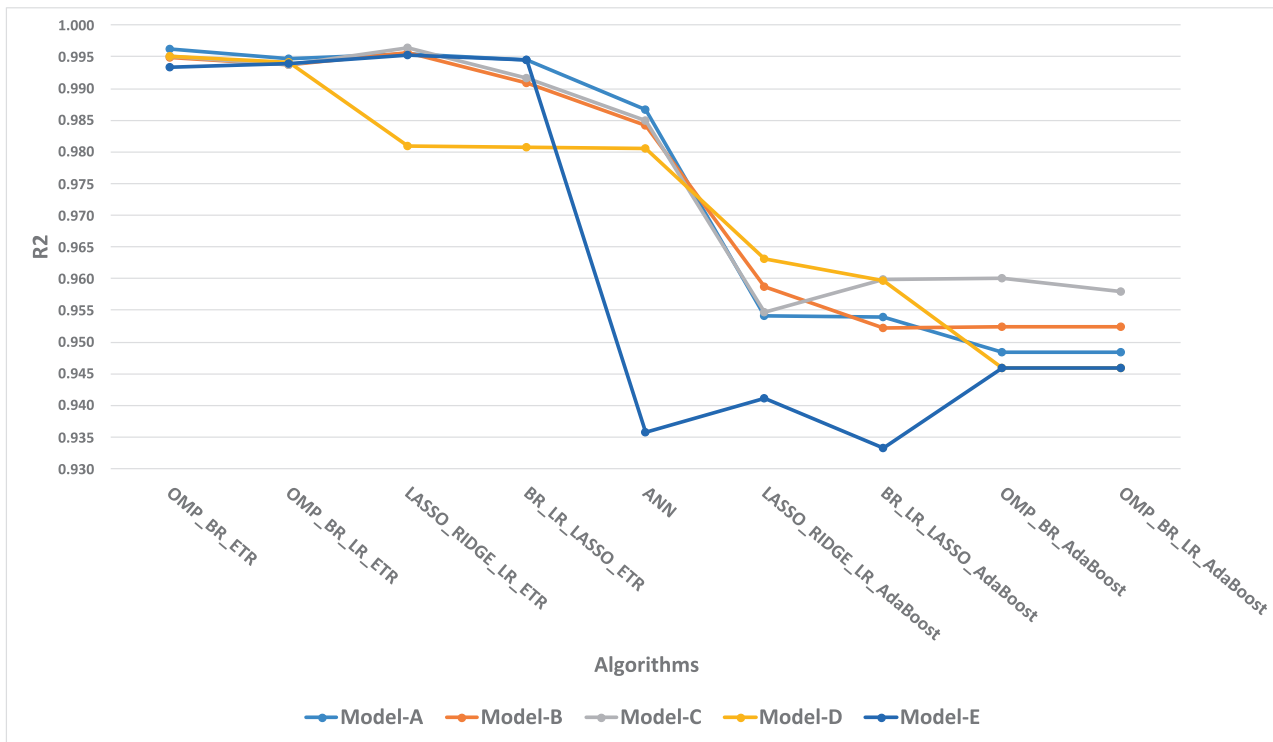


Fig. 8. Comparison between different models using R^2 .

and transferability across countries with diverse mobility patterns and infrastructure characteristics. The unusually low accuracy observed for the Slovak Republic is due to data-specific issues, rather than any limitation of the model. Collectively, these results highlight the power, reliability, and broad applicability of the proposed SEML-based Model-C approach for cross-country TED prediction.

6.3. Comparative performance analysis with country-specific stacking models

In the results section, we emphasized the impressive performance of our proposed SEML model. Remarkably, although it was not initially trained on datasets from Turkey, the model demonstrated exceptional

effectiveness in predicting the TED for the years 2018, 2019, and 2020. With an outstanding accuracy rate of 95.36%, this highlights its potential as a reliable forecasting tool across a range of contexts.

In the study conducted by Hoxha et al. [38], the authors examined the use of advanced stacking models with a specific dataset from Turkey. Their innovative approach integrated several base learners OMP, BR, and LR at the first level of their model, while employing ETR algorithm as the meta regressor at the second level. To evaluate and test the robustness of our framework, we implemented the same model but expanded our analysis to include a more extensive dataset from 14 different countries, excluding the Turkish dataset.

This comprehensive framework successfully advanced predictions for TED in Turkey for the years 2018, 2019, and 2020, achieving an

impressive accuracy rate of 95.38%. A detailed comparative analysis is presented in Table 10, which highlights the differences between the referenced study, which relied solely on data from Turkey, and our enhanced framework that leveraged a multi-national dataset. This comparison compellingly illustrates the improved strength, scalability, and generalization capabilities of our proposed framework, despite both frameworks using the same stacking configuration, with OMP, BR, and LR as base learners and ETR as the meta regressor.

Both models exhibited exceptional R^2 values exceeding 0.99, which attests to their predictive capabilities. Notably, the reference model achieved a slightly superior R^2 of 0.9955, while our proposed framework attained a commendable R^2 of 0.9946. This slight variation can be attributed to the shift from a context limited to a country-specific to a generalized learning paradigm that considers the diverse economic, demographic, and transportation patterns prevalent across multiple nations.

Significantly, our proposed framework achieved a remarkable reduction in both absolute and squared error metrics. The MAE and RMSE experienced a substantial decline, dropping from 288.7 and 335.3 to 0.8869 and 1.5243, respectively. Concurrently, the MSE decreased from an initial value of 112,433.8 to just 2.3235. These findings underscore a vastly more stable and precise predictive performance, even when faced with increased data heterogeneity across the diverse dataset.

Although the MAPE is higher in the multi-country framework due to the effects of scale normalization and inherent differences in demand magnitude across various countries, this trade-off is comfortably outweighed by the significant improvements in robustness and error reduction. In summary, our findings affirm that the proposed SEML not only maintains nearly optimal explanatory power but also surpasses the original model in terms of accuracy, stability, and scalability. This positions SEML as a highly effective and adaptable solution for forecasting TED across various national systems.

In a subsequent experiment designed to highlight the significance of generalization and the advantages of training a model with a diverse set of international data, we concentrated on utilizing data from Turkey to assess our framework performance in a different context. We implemented our proposed SEML framework, which integrates LASSO, RIDGE, and LR at the first level, with an ETR serving as the meta regressor specifically adapted for this dataset. The results were truly remarkable, showcasing an R^2 value of 0.9528, complemented by a MAPE of 4.1450, a MSE of 0.7796, a RMSE of 0.8830, and a MAE of 0.8735.

In Turkey, where our model was trained on historical data, it demonstrated exceptional performance by achieving an impressive accuracy rate of 98.02% in predicting the Turkish TED from the years 2018 to 2020. However, when we evaluated the capabilities in a wider context across 14 additional countries during the same timeframe, the results were significantly less encouraging. Poland emerged as the strongest performer among these nations with an accuracy of 89.10%, followed closely by Italy at 82.47% and the United Kingdom at 72.71%. The model's accuracy dwindled further with France, Australia, and Spain recording lower rates of 66.47%, 64.25%, and 62.82%, respectively. Alarming, for several remaining countries, the accuracy plummeted to below 45%.

Table 10

Comparison with the conducted research using Turkish data and our findings across multiple countries for SEML OMP, BR, and LR at the first level, with ETR as a meta regressor.

	Hoxha et al. [38] research	Our Research
R^2	0.9955	0.9946
MAPE	0.0303	11.0329
MSE	112 433.8	2.3235
RMSE	335.3110	1.5243
MAE	288.7170	0.8869

The results highlight a significant contrast between the model's strong correlation with training data from Turkey and its difficulties in generalizing and accurately predicting outcomes for other countries.

In Table 11, we present a comparison between our research and other recent studies. The comparison includes the country used in the research, the algorithm employed, the significant features considered, and the R^2 result. We are pleased to note that our model demonstrates a competitive R^2 value compared to the other results.

7. Conclusion

The comparative evaluation presented in this study confirms that the proposed global SEML model provides a robust and generalizable alternative to previously published country-specific TED forecasting approaches. Using Turkey as an out-of-sample test case, the results demonstrate that a model trained exclusively on multi-country data can accurately predict national TED without relying on local historical inputs. When compared with prior SEML-based studies focused solely on Turkey, the proposed framework delivers equally strong, and in some cases superior, predictive performance, underscoring its strong generalization capability and resistance to overfitting national trends. Moreover, while earlier studies typically limit their analysis to short or medium term horizons, this work successfully extends TED projections for Turkey up to 2040, providing added value for long-term transport and energy policy planning. These findings emphasize that global, reusable machine learning frameworks can effectively replace isolated country-level models, reducing data dependency while enhancing scalability, consistency, and applicability across diverse national contexts.

It is widely recognized that energy conservation and efficiency are significant challenges for countries worldwide. According to Eurostat's 2022 report, the transportation sector consumed the largest share of oil, accounting for 66% of total oil usage. In response to this issue, a comprehensive study was conducted to develop a framework capable of forecasting the energy requirements of the transportation sector across different countries. The study also aimed to identify the most crucial factors for accurate predictions. Unlike previous research, which predominantly focused on individual countries, this study analyzed groups of countries. It drew upon data from the OECD organization,

Table 11

Comparison between proposed framework and previous work.

Paper	The Countries data used	Algorithm	Features	R^2
(Sahraei and Çodur 2022) [42]	Turkey	ANN-PSO	Freight-km, Vehicle-km, Passenger-km, GDP Population, Oil-price	0.9931 Average R^2 : 0.9962
(Hoxha, et al. 2023) [38]	Turkey	SEML with OMP, BR, and LR in the first level, and ETR in the second level	Freight-km, Vehicle-km, Passenger-km, GDP Population, Oil-price	0.9955
(Korkmaz 2023) [29]	Turkey	BeSD	Freight-km, Passenger-km, Co2, GDP, Investment Per each transport mode (highway, railway, seaway, airway)	0.9969
Proposed work	14 countries	SEML with LASSO, RIDGE, and LR in the first level and ETR in the second level	TON-KM-RD, TON-KM-RL, P-KM-RD, P-KM-RL, GDP, POP	0.9963

encompassing various information such as transportation, socioeconomic, and global data. The study encompassed 14 countries, including Australia, Czech Republic, Finland, France, Hungary, Japan, Norway, Poland, Russia, Slovak Republic, Spain, Sweden, Switzerland, and the United Kingdom, which maintained complete records dating back to at least 2000. Utilizing data from 2000 to 2020, the study evaluated various feature sets with nine predictive algorithms. These included SEML of two levels including different combinations of OMP, BR, LR, LASSO, and Ridge Regressions for the first level, and ETR, and ADABOOST as meta regressors, along with ANN. Several metrics were employed to assess the models, including R2, MSE, RMSE, MAE, and MAPE. The best results were achieved by the SEML algorithm model which utilized LASSO, RIDGE, and LR for the first level, followed by ETR as the meta regressor for Model-C that consists of six features containing the transportation category except the investment features, and the socioeconomic category. The algorithm achieved R2 0. 0.9963, and values 14.6717, 1.5728, 1.2541, 0.7727 for MAPE, MSE, RMSE, MAE respectively. Also, this algorithm provides a significant result for Model-E which contains only three features (P –KM-RD, GDP, POP) with R2 0.9952, and values 7.7656, 2.0542, 1.4332, 0.9092 for MAPE, MSE, RMSE, MAE respectively. Notably, this study investigates Turkey's TED through two scenarios. The first scenario maintains the current distribution of demand across transportation modes, projecting the TED until 2040 based on existing trends. The second scenario reflects a shift from highways to alternative transportation modes, emphasizing their increasing popularity. Predictions indicate a demand of 41.05 MTOE for scenario 1 and 40.24 MTOE for scenario 2. We recommend implementing policies that favor alternative transportation modes such as public transit, cycling, and rail over highways to reduce energy consumption and promote sustainability in both passenger and freight transportation.

We strongly advocate for policymakers to adopt strategies that promote the development and utilization of alternative transportation modes over conventional highway-based systems for both passenger travel and freight transport. Prioritizing options such as public transit, cycling infrastructure, rail, waterway, and air transport can lead to substantial reductions in energy consumption within the transportation sector. This critical shift not only addresses growing energy demands but also supports a more sustainable and environmentally responsible approach to the movement of people and goods.

8. Limitations

Although this study demonstrates robust and significant results, certain boundaries should be noted. The model was developed using data from 14 countries, providing a strong foundation while leaving scope for further expansion to additional regions. Additionally, the analysis focused on two key transportation modes road and rail offering valuable insights while paving the way for future inclusion of other modes.

9. Future work

Building on the promising findings of this study, future research will concentrate on:

- Comparing our stacking-based framework with DL techniques to enhance prediction accuracy further.
- Developing optimized ANN model with optimization algorithms such as GA, and PSO
- Expanding the framework to encompass other energy types, such as electricity, to assess its generalizability.
- Conducting focused studies in countries like Egypt to aid in energy policy formulation and infrastructure development.

Ethical approval

This material is the authors' own original work, which has not been previously published elsewhere. The paper is not currently being considered for publication elsewhere. The paper reflects the authors' own research and analysis in a truthful and complete manner.

CRediT authorship contribution statement

Ahmed Samir: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Abeer ElKorany:** Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation, Conceptualization. **Manar ElKady:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eij.2026.100912>.

Data availability

The dataset is collected from the Organization for Economic Co-operation and Development (OECD) [49], and the U.S. Energy Information Administration (EIA) [50] for additional information, you could contact the corresponding author.

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